

Scalar vacuum polarization and the electric-field response in de Sitter space

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Abstract

We study the quadratic scalar effective action about a vanishing gauge background. For a Bunch–Davies scalar with $M_\phi^2 > 0$, the invariant vacuum-polarization kernel can be reconstructed from one scalar structure function. We give an explicit finite-part quadrature, including its subtraction scale and remaining local Maxwell term. Independently, expanding the exact Pauli–Villars current in a constant physical electric field gives a closed expression for its linear response. Cancelling this response defines a finite subtraction on one external history. Its identification with the form-factor condition in the project remains an additional matching problem. The local ultraviolet coefficient agrees with Step 2.

1 The quantity to be determined

The project [1] asks for the scalar-loop correction to the photon two-point function and a field-strength counterterm. The paper [2] uses a flat-space form factor, with a curvature shift in the scalar mass, to impose

$$\delta_3 = -e^2 \Pi(s_0), \quad s_0 = 2H^2 = -m_A^2. \quad (1)$$

The ultraviolet logarithm obtained this way is correct. The finite de Sitter calculation requires a specification of the kernel, external mode and boundary conditions.

Two calculations will be kept separate. First, we construct an invariant renormalized extension of the quadratic kernel with an explicit finite-part prescription. Within scalar QED power counting, its remaining bulk local freedom is an F^2 term. Second, we extract the linear response from the exact current already available in the paper. That current uses a specified Pauli–Villars regulator and an electric background prepared in the asymptotic past. It supplies a finite projection, but does not by itself fix the local matching between the two constructions.

Throughout,

$$ds^2 = a^2(\eta)(-d\eta^2 + d\mathbf{x}^2), \quad a = -\frac{1}{H\eta}, \quad M_\phi^2 = m_\phi^2 + 12\xi H^2 > 0, \quad \nu^2 = \frac{9}{4} - \frac{M_\phi^2}{H^2}. \quad (2)$$

The scalar in the quadratic kernel is in the Bunch–Davies state at $A = 0$. No internal photon propagator is needed: both lines in the loop are scalar. The restrictions on the tachyonic vector theory found in Step 3 matter when interpreting the external condition (1).

2 Current convention and the field equation

Let

$$S_\phi[A] = - \int d^4x \sqrt{-g} \left[g^{\mu\nu} (D_\mu \phi)^* D_\nu \phi + M_\phi^2 \phi^* \phi \right], \quad D_\mu = \nabla_\mu - ieA_\mu. \quad (3)$$

We use the *variational current*, placed on the left-hand side of the effective Maxwell equation:

$$\mathcal{J}^\mu = \frac{1}{\sqrt{-g}} \frac{\delta \Gamma_{\phi,L}^{(1)}}{\delta A_\mu}, \quad \mathcal{J}_\mu = -ie \langle \phi^* D_\mu \phi - (D_\mu \phi)^* \phi \rangle. \quad (4)$$

In real time this means the physical closed-time-path derivative with fixed initial density matrix. The source current on the right-hand side would be $J_{\text{src}} = -\mathcal{J}$. Expansion of (4) produces $-2e^2 A_\mu \langle \phi^* \phi \rangle$, the seagull contribution.

The classical vector action and counterterm give

$$(1 + \delta_3) \nabla_\nu F^{\nu\mu} - m_A^2 A^\mu + \mathcal{J}_{\text{reg}}^\mu = 0, \quad (5)$$

$$\mathcal{J}_{\text{ren}}^\mu = \mathcal{J}_{\text{reg}}^\mu + \delta_3 \nabla_\nu F^{\nu\mu}. \quad (6)$$

For the background of the project,

$$A_0 = 0, \quad A_z = \frac{E}{H^2 \eta} = -\frac{aE}{H}, \quad \nabla^\mu F_{\mu z} = -a^{-2} A_z'' = +2aHE. \quad (7)$$

Thus $\mathcal{J}_{\text{ct},z} = +2aHE\delta_3$. Moving the current to the right reverses both current terms and leaves a subtraction coefficient determined by their cancellation unchanged.

The paper uses the opposite sign for the charge in the covariant derivative. Relabelling $\phi \leftrightarrow \phi^*$ maps it to (3). Its tabulated current is denoted $\mathcal{J}$ below and used with (5). Retaining the tabulated sign while identifying it with $-\delta\Gamma/\delta A$ would be inconsistent.

3 The primitive two-point kernel

Write $G_F = \langle T\phi(x)\phi^*(x') \rangle$, so that

$$(\square_x - M_\phi^2)G_F(x, x') = i\delta_g(x, x'), \quad \delta_g(x, x') = \frac{\delta^4(x - x')}{\sqrt{-g(x)}}. \quad (8)$$

This is the object conventionally called $i\Delta$; it denotes the correlator itself. The closed-time-path components are

$$G_{-+} = G^+, \quad G_{+-} = G^-, \quad G_{++} = \theta(\eta - \eta')G^+ + \theta(\eta' - \eta)G^-. \quad (9)$$

The Bunch–Davies time-ordered correlator is the $++$ component. A matrix element between separately specified in and out vacua is a different object; no out vacuum is assumed here.

Let $K^{\mu\nu}$ be the density kernel given by the second functional derivative of the effective action. In diagrammatic notation $\mathcal{P}^{\mu\nu} = iK^{\mu\nu}$, the primitive Feynman expression is [3, 4]

$$\begin{aligned} \mathcal{P}^{\mu\nu}(x, x') = & -2ie^2 \sqrt{-g} g^{\mu\nu} G_F(x, x) \delta^4(x - x') \\ & + 2e^2 \sqrt{-g} g^{\mu\rho} \sqrt{-g'} g^{\nu\sigma} [\partial_\rho G_F \partial'_\sigma G_F - G_F \partial_\rho \partial'_\sigma G_F] \\ & + i\delta_3 \partial_\rho \left\{ \sqrt{-g} (g^{\mu\nu} g^{\rho\sigma} - g^{\mu\sigma} g^{\nu\rho}) \partial_\sigma \delta^4(x - x') \right\}. \end{aligned} \quad (10)$$

The lines are the seagull, derivative-vertex bubble and local counterterm. They are regulated distributions; products at coincidence cannot be assigned values using a separated-point formula alone. Define the undensitized kernel by

$$K^{\mu\nu}(x, x') = \sqrt{-g} g^{\mu\rho} \sqrt{-g'} g^{\nu\sigma} \Pi_{\rho\sigma'}(x, x'). \quad (11)$$

Then K is integrated with d^4x' , and Π with $d^4x' \sqrt{-g'}$. The diagrammatic object is i times this response convention.

At separated points the bubble divergence reduces to

$$\nabla^\rho(\nabla_\rho G \nabla_{\sigma'} G - G \nabla_\rho \nabla_{\sigma'} G) = (\square G) \nabla_{\sigma'} G - G \nabla_{\sigma'} \square G = 0. \quad (12)$$

At coincidence, (8) supplies delta functions cancelled by the seagull. A gauge-invariant regulator preserves this Ward identity. Density kernels obey ordinary divergences, $\partial_\mu K^{\mu\nu} = \partial'_\nu K^{\mu\nu} = 0$; the equivalent covariant statement is $\nabla^\rho \Pi_{\rho\sigma'} = 0$. The local logarithm must reproduce Step 2:

$$\delta_{3,\text{div}} = -\frac{e^2}{48\pi^2} \ln \frac{\Lambda^2}{\mu_R^2}, \quad \frac{\partial \delta_3}{\partial \ln \Lambda^2} = -\frac{e^2}{48\pi^2}. \quad (13)$$

4 An invariant finite-part kernel

4.1 Reduction to a scalar equation

Use $y = 2(1 - Z)$, with $0 < y < 4$ on the Euclidean section away from coincidence and the antipode. Step 1 supplies

$$T_\nu(y) = \frac{H^2}{16\pi^2} \Gamma\left(\frac{3}{2} + \nu\right) \Gamma\left(\frac{3}{2} - \nu\right) {}_2F_1\left(\frac{3}{2} + \nu, \frac{3}{2} - \nu; 2; 1 - \frac{y}{4}\right). \quad (14)$$

Define

$$\mathcal{K}_{\rho\sigma'}[\gamma] = \nabla^\alpha \nabla'^\beta \nabla_{[\rho} \nabla'_{\sigma']} \left\{ (\nabla_{\alpha]} \nabla'_{\beta]} y \right\} \gamma(y). \quad (15)$$

Each antisymmetrization has weight 1/2; the double bracket distinguishes the two index pairs. The reconstructed diagrammatic kernel is

$$\mathcal{P}_{\text{ren}}^{\mu\nu} = \sqrt{-g} g^{\mu\rho} \sqrt{-g'} g'^{\nu\sigma} \mathcal{K}_{\rho\sigma'}[\gamma_{\text{ren}}]. \quad (16)$$

Matching (10) away from coincidence gives [3]

$$\mathcal{D}_B \gamma = S_\nu, \quad \mathcal{D}_B = (4y - y^2) \frac{d^2}{dy^2} + 4(2 - y) \frac{d}{dy} - 2, \quad (17)$$

$$Q_\nu(y) := S'_\nu(y) = \frac{2e^2}{3H^4} \left\{ (4y - y^2) [(T'_\nu)^2 - T_\nu T''_\nu] - (2 - y) T_\nu T'_\nu \right\}. \quad (18)$$

Choose the primitive

$$S_\nu(y) = - \int_y^4 dz Q_\nu(z), \quad S_\nu(4) = 0. \quad (19)$$

A shift $S_\nu \mapsto S_\nu + C$ gives $\gamma \mapsto \gamma - C/2$, since $\mathcal{D}_B(1) = -2$. A constant γ is annihilated by $\mathcal{K}$, whose equivalent representation involves only γ' and γ'' . The primitive ambiguity therefore has no effect on the kernel. It is not a homogeneous solution of $\mathcal{D}_B$.

The homogeneous solutions are $B(y) = H^2/(4\pi^2 y)$ and $B(4 - y)$. The Green function with the corresponding regular endpoint choices is

$$G_B(y, z) = \begin{cases} -z/(4y), & 0 < z < y, \\ -(4 - z)/[4(4 - y)], & y < z < 4. \end{cases} \quad (20)$$

It is continuous, with derivative jump $1/[z(4 - z)]$, and satisfies $\mathcal{D}_{B,y} G_B(y, z) = \delta(y - z)$.

4.2 Endpoint subtraction

The short-distance expansion is

$$T_\nu(y) = \frac{H^2}{4\pi^2 y} + \frac{M_\phi^2 - 2H^2}{16\pi^2} \ln \frac{y}{4} + O(1). \quad (21)$$

It implies

$$S_\nu(y) = \frac{\mathfrak{a}}{y^2} + O\left(\frac{\ln y}{y}\right), \quad \mathfrak{a} = \frac{e^2}{24\pi^4}. \quad (22)$$

The primitive integral $\int_0^4 G_B S_\nu dz$ contains $-\mathfrak{a}(4y)^{-1} \int_0^y dz/z$. An explicit finite-part prescription is

$$\begin{aligned} \gamma_{\text{FP}}(y; y_*, c_R) = & -\frac{1}{4y} \left[\mathfrak{a} \ln \frac{y}{y_*} + \int_0^y dz \left(z S_\nu(z) - \frac{\mathfrak{a}}{z} \right) \right] \\ & - \frac{1}{4(4-y)} \int_y^4 dz (4-z) S_\nu(z) + \frac{c_R}{y}, \quad y_* > 0. \end{aligned} \quad (23)$$

Both integrals converge for fixed $0 < y < 4$. The first bracket is equivalently

$$\lim_{\varepsilon \rightarrow 0} \left[\int_\varepsilon^y z S_\nu(z) dz + \mathfrak{a} \ln \frac{\varepsilon}{y_*} \right]. \quad (24)$$

If U, V are the two numerators, $U' = y S_\nu$ and $V' = -(4-y) S_\nu$. Differentiating twice gives $\mathcal{D}_B \gamma_{\text{FP}} = S_\nu$. The second term tends to zero at $y = 4$, and the first remains finite. An independent $1/(4-y)$ term would introduce an antipodal singularity and is excluded.

To define the renormalized distribution, start on the Euclidean section, where the leading $\ln y/y$ singularity is locally integrable in four dimensions. Continue with the Bunch–Davies contour, take distributional boundary values, and apply all derivatives in (15) as distributional derivatives. Delta functions generated by differentiating B must be retained. This defines the contact extension together with the separated-point kernel. Within the local, covariant, gauge-invariant extension of scalar QED with its usual power counting, the remaining bulk local freedom is the Maxwell term c_R/y .

4.3 Scale and finite local matching

The contact identity in [3], Eqs. (85)–(86), fixes

$$\gamma_{\text{ct}} = -\frac{2\delta_3}{H^2} B(y) = -\frac{\delta_3}{2\pi^2 y}, \quad \Delta\delta_3 = -2\pi^2 \Delta c_R. \quad (25)$$

Writing $y_* = H^2/\mu_R^2$, a scale change at fixed c_R gives

$$\Delta\gamma_{\text{FP}} = \frac{\mathfrak{a}}{4y} \Delta \ln y_*, \quad \Delta\delta_3 = -\frac{e^2}{48\pi^2} \Delta \ln y_* = \frac{e^2}{48\pi^2} \Delta \ln \mu_R^2. \quad (26)$$

This agrees with Step 2. Scale changes and c_R shifts describe the same local freedom; fixing both specifies a scheme.

The choice $c_R = 0$ defines a finite-part reference scheme. Its finite normalization has not been equated with that of the paper’s Pauli–Villars current. The conversion is a local F^2 coefficient, obtainable by evaluating the same finite projection in both schemes. The Pauli–Villars subtraction coefficient below therefore cannot simply be inserted into (23).

4.4 Checks

For the conformal scalar, $T = H^2/(4\pi^2 y)$. Using the equivalent primitive $S = \mathfrak{a}/y^2$, Eq. (23) gives

$$\gamma_{\text{conf}}(y) = -\frac{\mathfrak{a}}{4y} \ln \frac{y}{y_*} - \frac{\mathfrak{a}}{4(4-y)} \left[\frac{4}{y} - 1 + \ln \frac{y}{4} \right] + \frac{c_R}{y}. \quad (27)$$

Direct differentiation gives $\mathcal{D}_B \gamma_{\text{conf}} = \mathfrak{a}/y^2$. The antipodal limit is finite:

$$\gamma_{\text{conf}}(4) = \frac{\mathfrak{a}}{16} \ln \frac{y_*}{4} + \frac{c_R}{4}. \quad (28)$$

Choosing $S(4) = 0$ instead adds the irrelevant constant $\mathfrak{a}/32$.

As a nonconformal check, take $M_\phi^2/H^2 = 1.25$, $H = e = 1$, $y_* = 1$, $c_R = 0$. High-precision quadrature at $y = 0.7, 2, 3.5$, followed by a five-point derivative with spacing 3×10^{-4} , gives $|\mathcal{D}_B \gamma - S| < 10^{-15}$. A numerical lower-endpoint cutoff changes the approximate answer only by a homogeneous $1/y$ term; its convergence was checked separately. These tests verify the scalar equation and subtraction, rather than the Lorentzian contact and current matching.

There is also an algebraic comparison with the dimensional subtraction in Ref. [3], Eqs. (97)–(98). Put $R = S_\nu - \mathfrak{a}/y^2$ and $\mathcal{G}[f] = \int_0^4 G_B(y, z) f(z) dz$. Its universal subtraction has the form

$$\gamma_{\text{DR}} = \frac{\mathfrak{a}}{4} \left[-\frac{\ln y}{y} + \frac{\ln(y/4)}{4-y} \right] + \mathcal{G} \left[R - \frac{\mathfrak{a}}{2y} \right]. \quad (29)$$

The residual integrals converge. Using

$$\mathcal{G}[1/y] = \frac{\ln(y/4)}{4-y} \quad \text{gives} \quad \gamma_{\text{DR}} = \gamma_{\text{FP}}(y; 1, \mathfrak{a}/4), \quad (30)$$

up to the irrelevant constant from the primitive. This checks the local normalization against that dimensional convention. It does not determine the finite conversion to the Pauli–Villars current.

5 Real-time and spatial representations

The contour separations are

$$y_{++} = aa' H^2 [r^2 - (|\eta - \eta'| - i0)^2], \quad (31)$$

$$y_{+-} = aa' H^2 [r^2 - (\eta - \eta' + i0)^2]. \quad (32)$$

The diagrammatic functions are $\gamma_{++} = \gamma(y_{++})$ and $\gamma_{+-} = -\gamma(y_{+-})$. Dividing the sum of the corresponding $\mathcal{P}$'s by i gives the real response kernel K_{ret} . It vanishes outside the past light cone. Local counterterms occur on the diagonal branch. These are relations between boundary values in the same state, not an identification of a retarded convolution with an independently defined in–out pole subtraction.

For smooth variations with controlled initial data,

$$\mathcal{J}_\mu^{(1)}(x) = \int d^4 x' \sqrt{-g'} \Pi_{\text{ret}, \mu\nu'}(x, x') A^{\nu'}(x'). \quad (33)$$

The spatial transform remains a two-time distribution:

$$\tilde{\Pi}_{\mu\nu'}(\eta, \eta'; \mathbf{k}) = \int d^3 r e^{-i\mathbf{k} \cdot \mathbf{r}} \Pi_{\mu\nu'}(\eta, \mathbf{x}; \eta', \mathbf{x} - \mathbf{r}). \quad (34)$$

Angular reduction uses

$$\int d\Omega e^{-i\mathbf{k}\cdot\mathbf{r}} = 4\pi j_0(kr), \quad (35)$$

$$\int d\Omega \hat{r}_i e^{-i\mathbf{k}\cdot\mathbf{r}} = -4\pi i j_1(kr) \hat{k}_i, \quad (36)$$

$$\int d\Omega \hat{r}_i \hat{r}_j e^{-i\mathbf{k}\cdot\mathbf{r}} = 4\pi \left[\frac{j_0(kr) + j_2(kr)}{3} \delta_{ij} - j_2(kr) \hat{k}_i \hat{k}_j \right]. \quad (37)$$

Equation (23) defines the distribution to be transformed. Differentiating only the separated-point bubble and treating it as an ordinary radial integrand loses the light-cone and coincident terms. The individual mixed components and their electric-history convolution have not been evaluated here.

6 The electric projection from the exact current

The current computed with Whittaker modes in the paper is a one-loop expectation value nonperturbative in the electric field. The same scalar determinant generates its Taylor coefficients about $A = 0$, provided the initial preparation is held fixed. In particular,

$$\left. \frac{\partial \mathcal{J}_z[A_E]}{\partial E} \right|_{E=0} = \int d^4x' \sqrt{-g'} \Pi_{\text{ret},z\nu'}(x, x') \frac{\partial A_E^{\nu'}(x')}{\partial E}. \quad (38)$$

The explicit gauge-field dependence of the current includes the seagull in this identity.

For the eternal Poincare-patch profile, the state is prepared in the asymptotic past with the positive-frequency prescription of the Whittaker modes. The weak-field limit below is an exact algebraic result for that current. Its identification with a convolution built from (23) also requires finite local matching and control of initial-surface terms as $\eta_i \rightarrow -\infty$. The external profile is neither spatially compact nor switched off in physical electric amplitude in the past. The identity for compact variations alone does not establish this preparation limit.

Define

$$\lambda = \frac{eE}{H^2}, \quad q = \frac{M_\phi^2}{H^2}, \quad \mu(\lambda)^2 = \frac{9}{4} - q - \lambda^2 = \nu^2 - \lambda^2. \quad (39)$$

The Pauli–Villars masses are $M_0^2 = M_\phi^2$, $M_1^2 = M_3^2 = 2\Lambda^2$, $M_2^2 = 4\Lambda^2 - M_\phi^2$, with weights $(-1)^i$. The regulated current [2] is

$$\mathcal{J}_{z,\text{reg}} = \frac{aHe^2E}{4\pi^2} \left[\frac{1}{6} \ln \frac{\Lambda^2}{H^2} - \frac{2\lambda^2}{15} + F_\phi(\lambda, \mu(\lambda)) \right]. \quad (40)$$

The full F_ϕ is recorded in Appendix A. We need the joint limit

$$F_0(\nu) = \lim_{\lambda \rightarrow 0} F_\phi(\lambda, \sqrt{\nu^2 - \lambda^2}). \quad (41)$$

7 Evaluation of the weak-field coefficient

For generic ν , put $s = \sin(2\pi\nu)$ and $A_0 = 45 + 8\pi^2(\nu^2 - 1)$. The first two terms of F_ϕ become

$$T_1 = \frac{A_0 + 4\pi^2\lambda^2}{12\pi^3} \frac{\mu \cosh(2\pi\lambda)}{\lambda^2 \sin(2\pi\mu)}, \quad (42)$$

$$T_2 = -\frac{A_0 + 64\pi^2\lambda^2}{24\pi^4} \frac{\mu \sinh(2\pi\lambda)}{\lambda^3 \sin(2\pi\mu)}. \quad (43)$$

Since

$$\frac{\mu}{\sin(2\pi\mu)} = \frac{\nu}{s} + \lambda^2 \left[\frac{\pi \cos(2\pi\nu)}{s^2} - \frac{1}{2\nu s} \right] + O(\lambda^4), \quad (44)$$

the $1/\lambda^2$ terms cancel. The derivative contribution in this expansion also cancels, leaving

$$\lim_{\lambda \rightarrow 0} (T_1 + T_2) = \frac{\nu(A_0 - 45)}{9\pi s} = \frac{8\pi\nu(\nu^2 - 1)}{9\sin(2\pi\nu)}. \quad (45)$$

In the integral term the polynomial becomes

$$P_0(r) = -1 + 4\nu^2 + (7 - 12\nu^2)r^2, \quad \int_{-1}^1 P_0(r) dr = \frac{8}{3}. \quad (46)$$

Writing $\psi_{\pm} = \psi(1/2 \pm \nu)$, the limit is

$$I_0 = \frac{1}{6} \operatorname{Re}[i \cot(\pi\nu)(\psi_+ - \psi_-) - (\psi_+ + \psi_-)]. \quad (47)$$

The reflection identity $\psi_+ - \psi_- = \pi \tan(\pi\nu)$ makes the first term purely imaginary, including for $\nu = i\rho$. Therefore

$$F_0(\nu) = \frac{8\pi\nu(\nu^2 - 1)}{9\sin(2\pi\nu)} - \frac{1}{6} \operatorname{Re} \left[\psi\left(\frac{1}{2} + \nu\right) + \psi\left(\frac{1}{2} - \nu\right) \right]. \quad (48)$$

The derivation assumes generic intermediate denominators. The final expression is defined by continuity at $\nu = 0, 1/2, 1$. The singularity at $\nu = 3/2$, corresponding to $M_\phi^2 = 0$, is not removable.

8 Two finite subtraction conditions

The linear current is

$$\mathcal{J}_{z,\text{reg}}^{(1)} = \frac{aHe^2E}{4\pi^2} \left[\frac{1}{6} \ln \frac{\Lambda^2}{H^2} + F_0(\nu) \right]. \quad (49)$$

In the local reference prescription of Step 2,

$$\delta_{3,\text{PV-MS}}(\mu_R) = -\frac{e^2}{48\pi^2} \ln \frac{\Lambda^2}{\mu_R^2}, \quad (50)$$

$$\mathcal{J}_{z,\text{PV-MS}}^{(1)} = \frac{aHe^2E}{4\pi^2} \left[\frac{1}{6} \ln \frac{\mu_R^2}{H^2} + F_0(\nu) \right]. \quad (51)$$

The finite de Sitter response remains in the current.

A different condition is

$$\left. \frac{\partial \mathcal{J}_{z,\text{ren}}[A_E]}{\partial E} \right|_{E=0} = 0. \quad (52)$$

Using $\mathcal{J}_{\text{ct},z} = 2aHE\delta_3$ fixes

$$\delta_{3,\text{proj}}^{\text{PV}} = -\frac{e^2}{8\pi^2} \left[\frac{1}{6} \ln \frac{\Lambda^2}{H^2} + F_0(\nu) \right]. \quad (53)$$

The finite difference from (50) is

$$\delta_{3,\text{proj}}^{\text{PV}} - \delta_{3,\text{PV-MS}}(\mu_R) = -\frac{e^2}{48\pi^2} \ln \frac{\mu_R^2}{H^2} - \frac{e^2}{8\pi^2} F_0(\nu). \quad (54)$$

This condition concerns one history in the state used for the current. For fixed $M_\phi^2 > 0$, its finite shift relative to the local prescription is well defined. It absorbs the infrared response of that mode as well as the local logarithm, and is not selected by ultraviolet renormalization.

In this scheme the full current is

$$\mathcal{J}_{z,\text{proj}} = \frac{aHe^2E}{4\pi^2} \left[-\frac{2\lambda^2}{15} + F_\phi(\lambda, \mu(\lambda)) - F_0(\nu) \right]. \quad (55)$$

Its analytic expansion starts at order E^3 because (52) was imposed. This is a property of the subtraction, not an independent prediction that the unsubtracted linear response vanishes. A finite F^2 shift in the effective action is the local renormalization freedom underlying these alternatives [6].

9 Mass limits and comparison with the paper

For a heavy scalar, $\nu = i\rho$,

$$F_0(i\rho) = -\frac{8\pi\rho(\rho^2 + 1)}{9\sinh(2\pi\rho)} - \frac{1}{3} \operatorname{Re} \psi\left(\frac{1}{2} + i\rho\right). \quad (56)$$

Thus $F_0 = -\frac{1}{3} \ln \rho + O(\rho^{-2})$, and

$$\delta_{3,\text{proj}}^{\text{PV}} = -\frac{e^2}{48\pi^2} \ln \frac{\Lambda^2}{H^2\rho^2} + O(e^2H^2/M_\phi^2), \quad H^2\rho^2 = M_\phi^2 - \frac{9}{4}H^2. \quad (57)$$

For $q = M_\phi^2/H^2 \ll 1$,

$$F_0 = \frac{3}{q} + O(1), \quad \mathcal{J}_z^{(1)} = \frac{3e^2aH^3E}{4\pi^2M_\phi^2} + O(q^0). \quad (58)$$

Since $A_z = -aE/H$, the leading response is $\mathcal{J}_z^{(1)} = -m_\gamma^2 A_z$, with

$$m_\gamma^2 = \frac{3e^2H^4}{4\pi^2M_\phi^2} + O(e^2H^2). \quad (59)$$

Placed on the left of (5), this has the sign of a positive induced mass squared, in agreement with [5]. It is infrared physics retained by (51) and absorbed on the chosen mode by (52). Neither choice removes the response of general modes. The exactly massless minimally coupled scalar needs a different infrared prescription and lies outside this invariant construction.

For a conformally coupled massless scalar, $q = 2$, $\nu = 1/2$, the poles in (48) cancel:

$$F_0(1/2) = \frac{1}{9} + \frac{\gamma_E}{3}, \quad \delta_{3,\text{proj}}^{\text{PV}} = \frac{e^2}{144\pi^2} \left[-3 \ln \frac{\Lambda^2}{H^2} - 2(1 + 3\gamma_E) \right]. \quad (60)$$

Write $\delta_3 = e^2[-3 \ln(\Lambda^2/H^2) + C(q)]/(144\pi^2)$. Then

$$C_{\text{proj}}(q) = -18F_0\left(\sqrt{\frac{9}{4} - q}\right), \quad (61)$$

$$C_{\text{flat}+R}(q) = 3 \ln q - 12q - 8 + 6(2q + 1)^{3/2} \operatorname{arccoth} \sqrt{2q + 1}. \quad (62)$$

At $q = 2$,

$$C_{\text{proj}}(2) = -5.463293989\dots, \quad C_{\text{flat}+R}(2) = 2.360112114\dots \quad (63)$$

The difference cannot yet be assigned to the neglect of curvature in the paper, because the finite conditions and external data also differ. A physical comparison requires the same scheme and the same response or spectral observable.

10 The original subtraction point and the remaining matching

Even in the flat approximation, subtracting a form-factor value differs from fixing a residue. In the convention implicit in (1),

$$D_T^{-1}(s) = s[1 + \delta_3 + e^2\Pi(s)] + m_A^2. \quad (64)$$

Setting $\delta_3 = -e^2\Pi(s_0)$, $s_0 = -m_A^2$, preserves the zero at s_0 , but

$$\left. \frac{dD_T^{-1}}{ds} \right|_{s_0} = 1 + e^2 s_0 \Pi'(s_0). \quad (65)$$

For $s_0 \neq 0$, a unit residue needs an additional condition and a compatible treatment of the mass and field normalization. The source's description as a pole-and-residue condition therefore does not itself define that stronger requirement. Subtracting the value of Π remains a possible prescription.

In exact de Sitter space a finite condition may instead be imposed on a specified harmonic or a retarded response. Step 3 explains why the homogeneous electric background does not automatically select a regular Euclidean vector harmonic or a positive-particle pole. Formal spectral continuation remains possible, but its continuation and subtraction data must be supplied.

The next direct calculation is now well defined: convolve the distribution from (23) with a regulated version of (7), keep the initial-surface terms, and match its finite local coefficient to the Pauli–Villars current before taking the preparation limit. Agreement with (49) would test the contact normalization, scheme conversion and asymptotic state together. The invariant finite-part kernel and algebraic weak-field coefficient have been obtained here; that convolution and the identification with the project's original condition have not.

A Exact current function and numerical check

The function in (40) is [2]

$$\begin{aligned} F_\phi(\lambda, \mu) = & \frac{45 + 4\pi^2(-2 + 3\lambda^2 + 2\mu^2)}{12\pi^3} \frac{\mu \cosh(2\pi\lambda)}{\lambda^2 \sin(2\pi\mu)} \\ & - \frac{45 + 8\pi^2(-1 + 9\lambda^2 + \mu^2)}{24\pi^4} \frac{\mu \sinh(2\pi\lambda)}{\lambda^3 \sin(2\pi\mu)} \\ & + \operatorname{Re} \int_{-1}^1 dr \frac{iP_\lambda(r)}{16 \sin(2\pi\mu)} \left\{ (e^{-2\pi r\lambda} + e^{2\pi i\mu}) \psi\left(\frac{1}{2} + \mu - ir\lambda\right) \right. \\ & \quad \left. - (e^{-2\pi r\lambda} + e^{-2\pi i\mu}) \psi\left(\frac{1}{2} - \mu - ir\lambda\right) \right\}, \end{aligned} \quad (66)$$

$$P_\lambda(r) = -1 + 4\mu^2 + (7 + 12\lambda^2 - 12\mu^2)r^2 - 20\lambda^2 r^4. \quad (67)$$

Direct high-precision evaluation at $\lambda = 0.005$, retaining $\mu^2 = \nu^2 - \lambda^2$, gives

ν	$F_0(\nu)$	$F_\phi(0.005, \mu)$	Difference
0.3	0.2407488614	0.2407529856	4.12×10^{-6}
0.7	0.4239111543	0.4239151013	3.95×10^{-6}
1.1	1.2255766518	1.2255729047	-3.75×10^{-6}
0.8i	0.0542532111	0.0542570944	3.88×10^{-6}
2i	-0.2275903364	-0.2275869313	3.41×10^{-6}

Halving λ reduces these differences by factors between 3.9998 and 4.0001, as expected for the next analytic term of order λ^2 . Near removable exceptional values, the combined expression or its series should be used instead of separately subtracting large poles.

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